Luis Gomez

(805) 314-4670 lgomezgonzalez@uiowa.edu https://lgomez010.github.io/

EDUCATION

Doctor of Philosophy Expected Dec 2028
Mathematics
University of Iowa, Iowa City IA

Master of Science Aug 2023-Dec 2025
Mathematics
University of Iowa, Iowa City IA

Bachelor of Science May 2020-May 2023
Mathematics
University of Arkansas, Fayetteville AR

TEACHING EXPERIENCE

University of Iowa Iowa City, IA
Instructor (College Algebra) Aug 2025-May 2026

- Lectured 30+ students on algebraic equations, techniques, inequalities, and functions

University of Iowa Iowa City, IA
Graduate Teaching Assistant Aug 2023-May 2025

- Led discussion sections of 40-160 students, answering questions and clarifying concepts

RESEARCH EXPERIENCE

University of Iowa, Iowa City, IA Mar 2026-Aug 2026

- Trained random forest classifiers to identify integers with 2 to 8 prime factors
- Built PySpark ML pipelines to process integer-vector encodings at scales up to 2^176

University of Arkansas, Fayetteville, AR Oct 2021-Dec 2022

- Analyzed structural properties of monomial ideals with Macaulay2
- Wrote code to compute minimal Gröbner bases of monomial ideals

Mathematical Sciences Research Institute, Berkeley, CA Jun 2022-Jul 2022

- Applied SageMath to study algebraic obstructions to convexity for neural codes
- Examined relationships between topological and algebraic definitions of neural codes

Rochester Institute of Technology, Rochester, NY Jun 2021-Jul 2021

- Characterized all 15 graphs with failed zero forcing number two
- Derived failed zero forcing numbers for circulant graphs on n vertices

SKILLS

Languages: Bash, LaTeX, Python

Libraries: Keras, Matplotlib, NumPy, Pandas, PyArrow, PySpark, Scikit-learn, Seaborn

Relevant coursework: Financial Mathematics (Stochastic Calculus), Numerical Methods 1, Advanced Numerical Methods (Fall 2026), Mathematics of Machine Learning, Analysis 1 and 2

PUBLICATIONS

- “**Random Forest Classification of r-Almost Primes**” with K. Wang, Y. Ye, (in preparation)
- “**Semiprime Detectability via Random Forests**” with K. Wang, Y. Ye, (in preparation)
- “**Failed Zero Forcing Number of Trees and Circulant Graphs**” with K. Rubi, J. Terrazas, D.A. Narayan, R. Flórez, *Theory and Applications of Graphs* (2025)
- “**All Graphs with a Failed Zero Forcing Number of Two**” with K. Rubi, J. Terrazas, D.A. Narayan, Symmetry, 13, 2221, 2021

TALKS

- Representation Theory Seminar, University of Iowa, Apr 10, 2026: “**Market hierarchies as p-adic spaces**”
- Representation Theory Seminar, University of Iowa, Feb 20 & 27, 2026: “**Automorphic representations and L-functions for GL(1) over Q**”
- Joint Mathematics Meetings, Jan 4, 2023: “**Wheel Signatures of Neural Codes**”
- Joint Mathematics Meetings, Apr 8, 2022: “**Failed Zero Forcing Graphs**”

EXTRACURRICULAR ACTIVITIES

Quantitative Finance Club — Iowa City, IA

Member — Aug 2025-Present

- Explore quantitative finance topics and network with peers across disciplines

LatinX Graduate Student Association — Iowa City, IA

Treasurer — Aug 2024-Dec 2025

- Managed budget and financial records, allocating funds across events

Bridge Program — Iowa City, IA

Member — Jan 2025-May 2025

- Completed a leadership program focused on intercultural communication and collaboration

RESEARCH INTEREST

- Computational and Probabilistic Number Theory

AWARDS

- Graduate Assistance in Areas of National Need (GAANN) Fellowship — Fall 2026
- Kleinfeld Endowment award — Fall 2023

CERTIFICATES

- Bloomberg Finance Fundamentals
- Center for the Integration of Research, Teaching and Learning (CIRTL) - Practitioner level
- DataCamp: Supervised Learning with scikit-learn, Unsupervised Learning in Python, Quantitative Risk Management in Python